

Effects of salinity on aerobic metabolism and development of eggs and alevins of steelhead trout (*Oncorhynchus mykiss*) and fall chinook salmon (*Oncorhynchus tshawytscha*)

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Metabolic rates, hatching success, alevin survival, time to hatch, and growth were measured for steelhead (*Oncorhynchus mykiss*) and fall chinook salmon (*Oncorhynchus tshawytscha*) embryos incubated in salinities of 0, 4, 8, and 12 parts per thousand (ppt) from the eyed stage. Metabolic rates of eyed steelhead trout eggs, and chinook salmon eggs and alevins were not significantly affected by the salinities tested. The metabolic rate of newly hatched steelhead trout alevins, however, was significantly lower in 8-ppt, and significantly higher in 12-ppt, than in 0- and 4-ppt salinities. Egg hatchability was high in all four salinities, but newly hatched alevins of both species showed decreased survival and were smaller in 12-ppt salinity. Salinity effects on alevin survival and size were greater for steelhead trout than for chinook salmon. The results suggest that 8-ppt salinity is the upper limit for the normal development of steelhead trout and chinook salmon eggs and alevins.

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Les taux de métabolisme, le succès de l'éclosion, la survie des alevins vésiculés, la durée de l'incubation et le taux de croissance ont été mesurés chez des embryons d'automne de Truites arc-en-ciel (*Oncorhynchus mykiss*) anadromes et de Saumons chinook (*Oncorhynchus tshawytscha*) gardés dans diverses conditions de salinité, soit 0, 4, 8 et 12 ppt, à partir du stade d'oeuf oculé. Les taux de métabolisme des oeufs oculés de Truites arc-en-ciel, des oeufs et des alevins vésiculés de Saumons chinook, n'étaient pas significativement affectés par les salinités expérimentales. Cependant, chez les alevins fraîchement éclos de la Truite arc-en-ciel, le taux de métabolisme était significativement plus faible à 8 ppt et significativement plus élevé à 12 ppt qu'aux salinités 0 et 4 ppt. L'éclosion des oeufs était élevée aux quatre salinités, mais les alevins fraîchement éclos des deux espèces avaient une servie moins grande et étaient plus petits à 12 ppt. Les effets de la salinité sur la survie des alevins vésiculés et sur leur taille étaient plus importants chez la Truite arc-en-ciel que chez le Saumon chinook. Ces résultats semblent indiquer qu'une salinité de 8 ppt constitue la limite supérieure du développement normal des oeufs et des alevins de la Truite arc-en-ciel et du Saumon chinook.

[Traduit par la rédaction]

Introduction

Differences in oxygen consumption rates in relation to salinity have been used extensively to estimate the energetic cost of ionic and osmotic regulation in juvenile and adult teleost fish (see Morgan and Iwama 1991). Although there have been several studies dealing with the effects of salinity on the metabolic rate of marine fish eggs and larvae (e.g., Lasker and Theilacker 1962; Holliday et al. 1964; Almatar 1984; Walsh and Lund 1989; Walsh et al. 1989), little is known about the influence of salinity on the rate of oxygen consumption in embryonic stages of anadromous salmonids. The primary aim of this study was to determine metabolic rates in eggs and alevins of steelhead trout (*Oncorhynchus mykiss*) and fall chinook salmon (*Oncorhynchus tshawytscha*) acclimated to four water salinities. As a secondary goal, we also examined the effects of salinity on survival, development time, and growth of steelhead trout and chinook salmon embryos.

Materials and methods

Egg acquisition and incubation

Steelhead trout and chinook salmon gametes were taken from fish returning to spawn in the Big Qualicum and Puntledge Rivers on Vancouver Island, B.C., in April and November, 1989, respectively. Steelhead trout eggs were pooled from three females and fertilized by the dry method with the pooled milt of three males. Water-hardened steelhead eggs had an average weight of 137 mg and an average diameter of 5.9 mm ($n = 15$). Chinook salmon eggs from a single Big Qualicum female were fertilized with milt from a single male. The eggs weighed 323 mg and were 8.1 mm in diameter ($n = 20$) after water-hardening.

The eggs sat in 1-mm nylon mesh baskets (11.5 cm diam., 2.5 cm high) near the top of 1.2-L cylindrical Perspex incubators (14 cm diam., 15 cm high; Rankin 1979). They were perfused with water entering at the bottom and passing upwards through a plastic cone. The incubators, therefore, provided a flat velocity front and micro-turbulent flow, perfusing one layer of eggs normal to the direction of flow ($300 \text{ mL} \cdot \text{min}^{-1} = 180 \text{ cm} \cdot \text{h}^{-1}$ perfusion velocity). Three

incubators were connected in series and were submerged in a 40-L incubation tank equipped with temperature control (Alderdice and Velsen 1968). Nominal water temperatures of 9 and 10°C were selected for incubation of steelhead and chinook eggs, respectively, and were kept within $\pm 0.1^\circ\text{C}$. Water circulation and aeration were maintained with an airstone across the back of the tank. Water was pumped from the bottom of the tank, through the incubators, and back into the tank. Four incubation tanks were set up to provide test salinities of 0, 4, 8, and 12 parts per thousand (ppt). Flow-through fresh and salt water were provided to each incubation tank from reservoirs, and flow rates were controlled by metered pumps to achieve the desired salinities (± 0.2 ppt).

About 250 newly fertilized eggs were placed in each incubator. The steelhead trout eggs were maintained in fresh water until after epiboly (117 accumulated thermal units or ATUs), whereas the chinook salmon eggs were kept in fresh water until after the eyed stage (299 ATUs). The eggs were kept in fresh water until these stages of development because water-hardening is known to be inhibited in salinities greater than 3 ppt (Li et al. 1989), and ion-regulating chloride cells do not appear until the basic structure of the embryo has fully formed (Alderdice 1988). The embryos were then gradually acclimated to the test salinities at a rate of 1–2 ppt/day. Salinity, temperature, and dissolved oxygen in each tank were recorded on a regular basis throughout the incubation periods (50 d for steelhead trout, 65 d for chinook salmon). Dissolved oxygen levels ranged from 10.4 to 11.3 $\text{mg} \cdot \text{L}^{-1}$ due to differences in oxygen solubility with increasing salt content, but were maintained in excess of 93% of air saturation. The minimum oxygen concentrations recorded were more than adequate to meet the dissolved oxygen requirements for the incubation of these species at the water temperatures and perfusion velocities used in the present study (Silver et al. 1963; Shumway et al. 1964; Davis 1975; Rombough 1986, 1988a).

All incubators were kept in darkness except during brief examination periods. Samples of approximately 55 eggs per salinity treatment, without replication, were set aside to determine the effect of salinity on hatching success and time to hatch. The incubators were opened every 3–4 d and any dead (opaque) eggs were removed and preserved in Stockard's solution to determine the embryonic stage at death (Ballard 1973). During hatching, the numbers of alevins that appeared were recorded daily, and alevin survival was monitored for about 1 week post-hatching. Hatching success was calculated as the percentage of eggs that survived from the beginning of salinity exposure to hatch. The time of hatching was taken as the number of days from fertilization to the time by which 50% of the eggs had hatched (Murray and McPhail 1988).

Respirometry

Routine metabolic rates of eyed eggs and newly hatched alevins incubated in the different salinity treatments were determined by monitoring rates of oxygen consumption in continuous-flow respirometers (Rombough 1988b). The respirometers were 71-mL cylindrical acrylic chambers, with nylon screens at either end. The chambers were coated with water-soluble silicone (Clay Adams Co., Parsippany, N.J.) to inhibit bubble formation. Stainless steel tubing was used to connect the chambers to dissolved-oxygen probe holders equipped with magnetic stirrers. A variable-speed gear pump and a four-way valve allowed the respirometer to operate in an open (flow through) or closed mode. Four respirometers were immersed in a common 40-L bath and regulated to $\pm 0.1^\circ\text{C}$ by an attached temperature control unit. Water temperatures for metabolic rate measurements were kept similar to those used in incubation (9.0–9.1°C for steelhead, 10.0–10.1°C for chinook). The bath was filled with appropriate volumes of fresh and salt water to achieve the test salinities, and the dissolved oxygen concentration of the bath water was adjusted to saturation by aeration in the temperature control unit. Dissolved oxygen levels in each chamber were monitored continuously using polarographic oxygen electrodes (calibrated with air-saturated distilled water) and a multichannel oxygen meter (Orbisphere Model 2710).

Metabolic rates were determined at the eyed stage (steelhead trout: 240–252 ATUs, chinook salmon: 429–443 ATUs) and approximately 1 week post-hatch (steelhead: 438–450 ATUs, chinook: 640–653 ATUs). Only normal, vigorous alevins were used. Six replicate trials were conducted per salinity treatment at each stage tested. Two days were required to complete the respiration measurements for each developmental stage. Depending on the species and stage of development, between 25 and 100 individuals were placed into each of three respirometers and the flow rate past the eggs or alevins was adjusted to 60 $\text{mL} \cdot \text{min}^{-1}$ (equivalent to a bulk water velocity of 510 $\text{cm} \cdot \text{h}^{-1}$). The fourth respirometer was run without animals to serve as a control. The top of the respirometer bath was covered with black plastic sheeting and the eggs or alevins were acclimated to the respirometers in flow-through conditions for 30 min. The respirometers were then closed and the subsequent decline in oxygen concentrations (to the nearest 0.05 $\text{mg} \cdot \text{L}^{-1}$) was monitored for 1 h, with values recorded every 5 min. One replicate during each respiration trial was plotted continuously, using a strip chart recorder, for comparison with the manual readings. After a trial was completed, 10 individuals from each respirometer were preserved in 10% neutral Formalin. After at least 120 d of preservation, the egg and alevin samples were measured for fork length and weight. Alevin fork length was measured to the nearest millimetre. The yolk and tissue components of eggs and alevins were dissected, and 10 individuals from each replicate were pooled before weighing. Preservation in Formalin is known to alter the weight of fishes (Heming and Preston 1981), and evaporation of Formalin during the dissection procedure also affected wet-weight measurements. Because of these problems associated with using preserved wet weights, only dry weights were determined. Tissue and yolk were oven-dried at 60°C for 48 h and weighed to the nearest 0.1 mg. Percent embryonic tissue (P_T) was calculated for each development stage by dividing tissue dry weight (TDW) by total egg or alevin dry weight (Rombough 1985). Gross conversion efficiencies were calculated as the gain in TDW divided by the loss in yolk dry weight (YDW) multiplied by 100 (Blaxter 1969).

Dissolved oxygen concentrations in the respirometers initially declined in a curvilinear fashion for 10–15 min, apparently as a result of pressure changes associated with closing the system (Rombough 1988b). Oxygen levels then decreased linearly for 30–45 min, during which time dissolved oxygen dropped 2–4 $\text{mg} \cdot \text{L}^{-1}$. Oxygen consumption rates were estimated over the linear portion by regression analysis (r^2 : 96.9–99.9%). Metabolic rates were determined by multiplying the regression slope of oxygen uptake (corrected for the blank controls) by the respirometer volume and were expressed in terms of tissue dry weights over a 1-h period (i.e., $\mu\text{g O}_2 \cdot \text{g TDW}^{-1} \cdot \text{h}^{-1}$).

Chloride content of the perivitelline fluid

Samples of the perivitelline fluids (PVF) were taken from 4–5 chinook salmon eggs from each of the four salinity treatments. These eggs were well eyed (420 ATUs) and had been exposed to the test salinities for about 5 days. The eggs were gently rolled onto tissue paper to remove surface moisture and were then mounted into small plastic cups. PVF was sampled by making a small hole in the chorion with a fine-tipped needle and withdrawing the fluid into 10- μL capillary tubes (Microcap®, Drummond Scientific Co., Broomall, Pa.). Chloride ion concentrations ($[\text{Cl}^-]$) were measured immediately by coulometric titration (Haake Buchler Instruments Digital Chloridometer). Appropriate reference standards were run to ensure measurement accuracy. Water samples from each incubation tank were also measured for $[\text{Cl}^-]$ for comparison with the PVF values.

Statistical analysis

Data are presented as means $\pm$ standard error (SE) where appropriate. The P_T values were normalized using the arcsine transformation prior to statistical analysis. The metabolic-rate and size data were subjected to analysis of variance. When significant effects were detected, Tukey's multiple-comparison test (Steel and Torrie 1980)

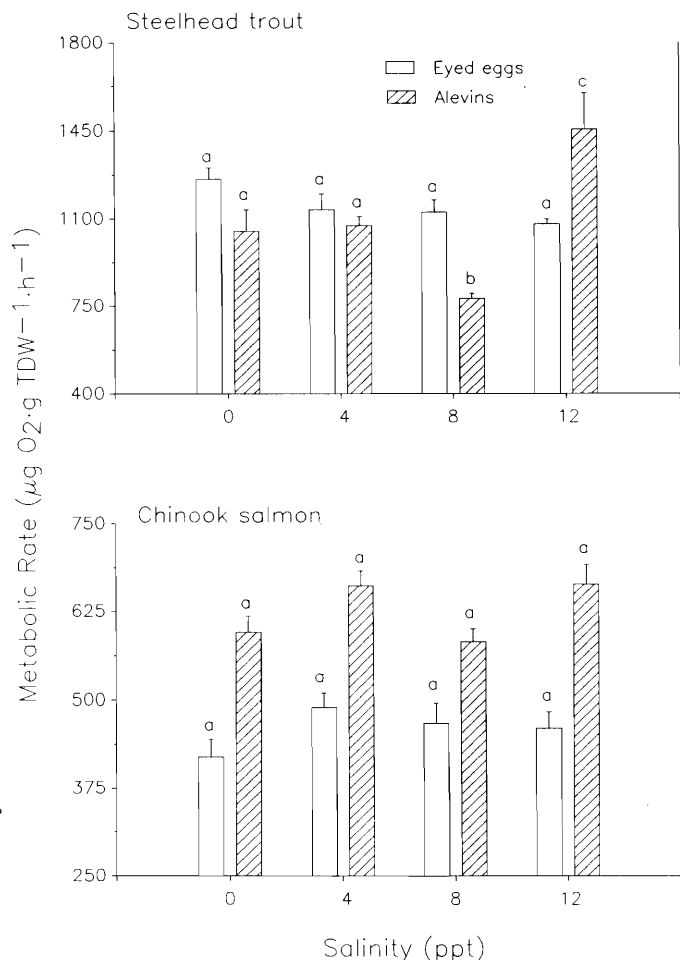


FIG. 1. Metabolic rates of eyed eggs and newly hatched alevins of steelhead trout and chinook salmon in four salinities. Means (+SE) with a common letter are not significantly different ($P < 0.05$) by Tukey's test.

was used to identify significantly different treatment means ($P < 0.05$). The $[Cl^-]$ determined for chinook egg PVF and incubation water in similar salinity treatments were compared using unpaired t -tests ($P < 0.05$). All analyses were performed with the SYSTAT statistical program (Wilkinson 1988).

Results

Metabolic rates

Routine metabolic rates ($\mu g O_2 \cdot g TDW^{-1} \cdot h^{-1}$) of eyed steelhead trout eggs and chinook salmon eggs and alevins were not significantly affected ($P < 0.05$) by incubating salinities of up to 12 ppt (Fig. 1). Metabolic rates of newly hatched steelhead alevins, however, were significantly lower in 8-ppt, and significantly higher in 12-ppt salinity, compared with the other salinity treatments.

Perivitelline fluid chloride content

There were no significant differences ($P > 0.05$) found between the $[Cl^-]$ of the PVF in eyed chinook salmon eggs and the incubation water for any given salinity treatment (Table 1).

Survival and development

Steelhead trout embryo survival from the beginning of salinity exposure to hatching was $>96\%$ in salinities of 0, 4, and 8 ppt but declined 15% in 12 ppt (Table 2). Most dead eggs

TABLE 1. Chloride content of the perivitelline fluid in eyed chinook salmon eggs acclimated to four salinities

Salinity (ppt)	Chloride (mequiv. $\cdot L^{-1}$)					
	Perivitelline fluid			Incubation water		
	$\bar{x}$	SE	<i>n</i>	$\bar{x}$	SE	<i>n</i>
0	8.2	2.3	4	2.9	0.5	3
4	57.6	3.8	5	62.5	0.5	2
8	123.6	2.7	5	123.0	5.0	2
12	185.8	4.9	5	190.0	2.0	2

NOTE: Unpaired t -tests between the chloride content of the PVF and the incubation water at similar salinities revealed no significant differences ($P > 0.05$).

TABLE 2. Survival and development times for steelhead trout and chinook salmon embryos incubated in four water salinities

Salinity (ppt)	Percent hatch	Time to 50% hatch (d)	7-d alevin survival (%)
Steelhead trout			
0	98.1	39	98.0
4	98.1	39	98.1
8	96.6	40	100
12	84.4	40	25.0
Chinook salmon			
0	100	53	100
4	100	53	100
8	98.0	54	100
12	98.1	53	90.4

NOTE: $n = 55$ eggs per salinity treatment, without replication.

in 12-ppt salinity were 'well eyed' (Stage 22b, Ballard 1973) and died just before hatching. Hatching success of chinook salmon eggs was $\geq 98\%$ in all four salinity treatments. Times to 50% hatching for steelhead trout and chinook salmon embryos in fresh water were similar to those predicted by models relating temperature and incubation time of Pacific salmonids (Alderice and Velsen 1978; Rombough 1986; Jensen 1988; Beacham and Murray 1990) and were not affected by incubating salinities of up to 12 ppt. The 7-d posthatch survival of steelhead trout and chinook salmon alevins was $>98\%$ in all salinities except 12 ppt, in which there was a 10% decline in chinook and a 75% decline in steelhead alevin survival (Table 2).

Egg and alevin size

Total dry weights of steelhead and chinook eggs did not differ significantly ($P < 0.05$) among the four salinity treatments (Table 3). The slight increase in P_T values with salinity reflected the chronological order of sampling, as eggs in 8- and 12-ppt salinities were sampled 1 d later than those in 0- and 4-ppt salinities.

One-week-old steelhead trout alevins had similar total dry weights in all salinities; however, the P_T values decreased with increasing salinity and were significantly lower ($P < 0.05$) in 8- and 12-ppt than in 0- and 4-ppt salinities. Steelhead alevin fork lengths also decreased significantly in 8- and 12-ppt salinities, reflecting the lower amounts of tissue in these salinity treatments. P_T values and fork lengths of chinook alevins were significantly lower ($P < 0.05$) only in 12-ppt salinity.

TABLE 3. Weights and lengths for eyed eggs and newly hatched alevins of steelhead trout and chinook salmon incubated in four salinities

Stage*	Salinity (ppt)	Total egg or alevin dry weight [†] (mg)	Tissue dry weight (mg)	Yolk dry weight (mg)	P_T	Gross conversion efficiency	Length (mm)
Steelhead trout							
Eyed eggs (640–672 h)	0	44.6±0.6a	1.5±0.0a	43.1±0.6a	3.4±0.1a		
	4	43.8±0.6a	1.6±0.1ab	42.2±0.6a	3.6±0.2ab		
	8	43.9±0.6a	1.7±0.0b	42.2±0.6a	3.9±0.1bc		
	12	43.1±0.3a	1.8±0.1b	41.3±0.3a	4.2±0.1c		
Alevins (1168–1200 h)	0	38.9±0.5a	14.0±0.1a	24.9±0.5a	36.0±0.5a	68.7	22.0±0.1a
	4	40.9±0.6a	14.2±0.2a	26.7±0.6ab	34.7±0.6a	81.3	21.8±0.1a
	8	39.8±0.5a	11.3±0.2b	28.5±0.3b	28.4±0.3b	70.1	20.7±0.1b
	12	45.4±1.9a	8.4±0.8c	37.0±1.0c	18.5±1.1c	153.5	19.0±0.2c
Chinook salmon							
Eyed eggs (1030–1063 h)	0	110.2±0.4a	8.4±0.1ab	101.8±0.4a	7.6±0.1ab		
	4	110.8±0.4a	8.1±0.5a	102.7±0.8a	7.3±0.5a		
	8	110.0±0.3a	8.4±0.2ab	101.6±0.4a	7.6±0.2ab		
	12	110.5±0.3a	9.2±0.1b	101.3±0.3a	8.3±0.1b		
Alevins (1536–1567 h)	0	99.9±0.5ab	26.8±0.3a	73.1±0.4a	26.8±0.2a	64.1	26.7±0.1a
	4	101.5±0.4a	28.7±0.2b	72.8±0.5a	28.3±0.2b	68.9	26.7±0.1a
	8	98.8±0.6b	26.6±0.4a	72.2±0.8a	26.9±0.5a	61.9	28.1±0.2b
	12	101.7±0.3a	24.2±0.4c	77.5±0.4b	23.8±0.3c	63.0	25.5±0.1c

NOTE: Values are given as the mean ± SE. For each species and stage, mean values in a column with a common letter are not significantly different ($P < 0.05$) by Tukey's test. For weights, $n = 6$ (except for steelhead alevins in 12-ppt salinity, for which $n = 2$), with each replicate representing a pooled average of 10 individuals. For lengths, $n = 20$ alevins per salinity treatment.

*Elapsed time is in parentheses.

[†]Dechorionated eggs.

Gross conversion efficiencies calculated for the period between eyed eggs and one-week-old alevins were generally similar at all salinities (61.9–81.3%), with the exception of the one for steelhead trout in 12-ppt salinity, which was 153.5% (Table 3). This erroneously high value may have been due to the intrusion of salt in the yolk of steelhead trout alevins in the hypertonic treatment.

Discussion

The metabolic rates determined for steelhead and chinook eggs and alevins in fresh water were comparable to values obtained by other workers for various salmonid species (Hayes et al. 1951; Alderdice et al. 1958; Carrick 1981; Wieser et al. 1985). Difficulties in comparing literature values arise from differences in the stages of development, activity levels, temperatures, and measurement techniques employed by different investigators. These factors can have large influences on the metabolic rates of eggs and alevins (Rombough 1988c). Rombough (1988a, 1988b, 1988c) conducted comprehensive studies on the aerobic metabolism and dissolved oxygen requirements of eggs and alevins of steelhead trout and chinook salmon, using the same apparatus and methodology employed in the present study. His predictive regression equations relating metabolic rates with incubation time and temperature indicated similar values for steelhead trout eggs and alevins to ours, whereas predicted metabolic rates for chinook salmon eggs and alevins were higher than those determined in this study. The reasons for the large (100%) difference between the metabolic rates recorded in the present study for steelhead trout and chinook salmon eggs and alevins are not known.

Metabolic rates of steelhead trout and chinook salmon eggs were unaffected by constant salinities of up to 12 ppt. These results are consistent with previous studies in which marine teleost embryos were used. Lasker and Theilacker (1962; Pacific sardine), Holliday et al. (1964; Atlantic herring),

Walsh and Lund (1989; cottids), and Walsh et al. (1989; striped mullet) reported no significant differences in metabolic rates of marine fish eggs and larvae incubated in various salinities. The present study is the first to record a similar lack of metabolic response to salinity for embryos of anadromous salmonid species. It is not clear from these data whether energetic costs for ion regulation are negligible when compared with total metabolic rate, or if increased ion regulatory costs are compensated for by decreases in other metabolic processes (Holliday et al. 1964). Alderdice (1988) suggests that internal ionic and osmotic balance in salmonid embryos is maintained mostly by the 'passive' regulation of ion concentrations by impermeable membranes surrounding the yolk and body tissues, rather than by energy-consuming mechanisms of active ionic and osmotic regulation. The absence of a salinity effect on metabolic rates of steelhead trout and chinook salmon eggs in the present study supports this hypothesis.

Metabolic rates of chinook salmon alevins were not affected by salinities of up to 12 ppt, whereas respiration rates measured for steelhead trout alevins were depressed in 8-ppt and elevated in 12-ppt salinities. It is difficult from the present data to distinguish metabolic costs associated with ionic and osmotic regulation in the overall context of routine metabolism. Recent work by Wieser et al. (1988) suggests that in larval fish, which must operate on fixed energy budgets, energy partitioning may be compensatory rather than additive. In other words, any increase in ion regulation costs may be masked by a concomitant decrease in some other metabolic cost, such as neuromuscular activity. Almatar (1984) found that after an abrupt change in salinity, herring and plaice yolk-sac larvae consumed more oxygen in seawater (32 ppt) than in low salinities (5 and 12.7 ppt). Although activity of the larvae was not examined, Almatar (1984) attributed the differences in oxygen uptake to possible changes in activity level in response to the salinity change. Activity levels of the salmonid alevins in the

respirometers could not be measured in the present study, making the results difficult to interpret in this regard. It is possible that the increased rate of oxygen uptake observed for steelhead trout alevins in 12-ppt salinity may have reflected a metabolic response to osmotic stress, rather than an energetic cost of ion regulation.

The $[Cl^-]$ of the PVF in eyed chinook salmon eggs closely matched the $[Cl^-]$ of the external medium. The chorion of salmon eggs has been shown to be permeable to sodium and chloride ions (Rudy and Potts 1969; Peterson and Martin-Robichaud 1987) and water (Potts and Rudy 1969). The perivitelline colloid has a net negative charge at normal pH levels (Peterson 1984) and can accumulate sodium ions at concentrations in excess of those present in fresh water (Rudy and Potts 1969). There were no indications in this study of significant accumulation of chloride ions in the PVF relative to the external medium. This was to be expected, since the perivitelline colloids would provide an anion excess and attract cations rather than anions. There have been suggestions in the literature that the chorion and PVF may afford some degree of protection to the developing embryo from adverse environmental conditions, such as acidic water, which is an advantage not available to dechorionated eggs and newly hatched alevins (Daye and Garside 1979; Eddy and Talbot 1985). Weisbart (1968) further showed that dechorionated eggs of all five species of Pacific salmon had slightly decreased survival times in seawater, compared with intact embryos. In this study, alevins in 12-ppt salinity showed an increase in mortality rates soon after hatching, which also suggests a decreased resistance to salinity, compared with eyed eggs. Why the chorion would impart increased salinity resistance to the embryo is unclear, since, as mentioned above, the chorion is permeable to external salt and the PVF has an osmotic pressure similar to that of the external environment (Shen and Leatherland 1978; Groot 1989). The role of the chorion and the PVF in the salinity tolerance of salmonid embryos clearly warrants further study.

The survival data suggest that a salinity of about 8 ppt is the upper limit for the normal development of steelhead trout and chinook salmon eggs and alevins. Eight parts per thousand is approximately isotonic with the blood of salmon alevins (Weisbart 1968), which suggests that continuous exposures to hypertonic salinities are not tolerated. Previous studies on the salinity tolerance of salmonid eggs and alevins have also shown decreased survival and increased abnormalities in salinities above the isotonic level (Saunders 1966; Leatherland and Lin 1975; Shen and Leatherland 1978; Veltishcheva 1983; Groot 1989). In our study, chinook salmon eggs and alevins had higher survival rates in 12-ppt salinity than were obtained for steelhead trout. This greater salinity resistance may have been due, in part, to a higher tolerance of chinook salmon for increased salinity at the tissue level, as proposed by Weisbart (1968).

The effects of water salinity on size differences of hatching larvae have been reported by many workers (see reviews by Holliday 1969; von Westernhagen 1988). In this study, steelhead and chinook alevins hatched in 12-ppt salinity were smaller than those hatched in lower salinities. However, there was no indication of decreased gross conversion efficiencies at the higher salinity, which suggests that the hypertonic salinity may have slowed the rate of yolk utilization rather than lowering its conversion efficiency to body tissue.

The results of the present study may have a practical application in aquaculture. Salt has recently become a popular alter-

native to Malachite Green as a fungicide for the culture of salmon eggs. Current hatchery practice involves the immersion of eggs in a static, 20-ppt salinity bath for 1 h, 3 times per week from water-hardening to the eyed stage (P. Edgell, Robertson Creek Hatchery, Port Alberni, B.C., personal communication). These data showed that constant incubation of steelhead trout and chinook salmon embryos in 4-ppt salinity following epiboly did not affect survival, rate of development, growth, or metabolism. Coastal hatcheries equipped with salt-water lines could therefore incubate salmonid eggs and alevins at slightly higher salinities (i.e., <5 ppt), which would probably decrease the incidence of infections by stenohaline freshwater bacteria and fungi, and may increase the salinity tolerance of emerging fry (Alderdice 1988). However, it should be pointed out that the risk of vibriosis may be increased in brackish water (Gjedrem and Aulstad 1974); the incoming water supply should therefore be screened for *Vibrio anguillarum*. As the present experiments were terminated when the alevins were only 1 week old, further testing would be necessary before implementation of such a strategy to ensure that the alevins were not affected in later stages of development.

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- Alderdice, D. F. 1988. Osmotic and ionic regulation in teleost eggs and larvae. In *Fish physiology*. Vol. 11A. Edited by W. S. Hoar and D. J. Randall. Academic Press, San Diego. pp. 163–251.
- Alderdice, D. F., and Velsen, F. P. J. 1968. Design of a controlled-environment incubator for small marine fish eggs. *J. Fish. Res. Board Can.* **25**: 585–588.
- Alderdice, D. F., and Velsen, F. P. J. 1978. Relation between temperature and incubation time for eggs of chinook salmon (*Oncorhynchus tshawytscha*). *J. Fish. Res. Board Can.* **35**: 69–75.
- Alderdice, D. F., Wickett, W. P., and Brett, J. R. 1958. Some effects of temporary exposure to low dissolved oxygen levels on Pacific salmon eggs. *J. Fish. Res. Board Can.* **15**: 229–250.
- Almatar, S. M. 1984. Effects of acute changes in temperature and salinity on the oxygen uptake of larvae of herring (*Clupea harengus*) and plaice (*Pleuronectes platessa*). *Mar. Biol. (Berlin)*, **80**: 117–124.
- Ballard, W. W. 1973. Normal embryonic stages for salmonid fishes, based on *Salmo gairdneri* and *Salvelinus fontinalis*. *J. Exp. Zool.* **184**: 7–25.
- Beacham, T. D., and Murray, C. B. 1990. Temperature, egg size, and development of embryos and alevins of five species of Pacific salmon: a comparative analysis. *Trans. Am. Fish. Soc.* **119**: 927–945.
- Blaxter, J. H. S. 1969. Development: eggs and larvae. In *Fish physiology*. Vol. 3. Edited by W. S. Hoar and D. J. Randall. Academic Press, New York. pp. 177–252.
- Carrick, T. R. 1981. Oxygen consumption in the fry of brown trout (*Salmo trutta*) related to pH of the water. *J. Fish Biol.* **18**: 73–80.
- Davis, J. C. 1975. Minimal dissolved oxygen requirements of aquatic life with emphasis on Canadian species: a review. *J. Fish. Res. Board Can.* **32**: 2295–2332.
- Daye, P. G., and Garside, E. T. 1979. Development and survival of embryos and alevins of the Atlantic salmon, *Salmo salar* L., continuously exposed to acidic levels of pH, from fertilization. *Can. J. Zool.* **57**: 1713–1718.

- Eddy, F. B., and Talbot, C. 1985. Sodium balance in eggs and dechorionated embryos of the Atlantic salmon, *Salmo salar*, exposed to zinc, aluminum and acidic waters. *Comp. Biochem. Physiol. C*, **81**: 259–266.
- Gjedrem, T., and Aulstad, D. 1974. Selection experiments with salmon. I. Differences in resistance to vibrio disease of salmon parr (*Salmo salar*). *Aquaculture*, **3**: 51–59.
- Groot, E. P. 1989. Intertidal spawning of chum salmon: saltwater tolerance of the early life stages to actual and simulated intertidal conditions. M.Sc. thesis, Department of Zoology, University of British Columbia, Vancouver, B.C.
- Hayes, F. R., Wilmot, I. R., and Livingstone, D. A. 1951. The oxygen consumption of the salmon egg in relation to development and activity. *J. Exp. Zool.* **116**: 377–395.
- Heming, T. A., and Preston, R. P. 1981. Differential effect of formalin preservation on yolk and tissue of young chinook salmon (*Oncorhynchus tshawytscha* Walbaum). *Can. J. Zool.* **59**: 1608–1611.
- Holliday, F. G. T. 1969. The effects of salinity on the eggs and larvae of teleosts. In *Fish physiology*. Vol. 1. Edited by W. S. Hoar and D. J. Randall. Academic Press, New York. pp. 293–311.
- Holliday, F. G. T., Blaxter, J. H. S., and Lasker, R. 1964. Oxygen uptake of developing eggs and larvae of the herring (*Clupea harengus*). *J. Mar. Biol. Assoc. U.K.* **44**: 711–723.
- Jensen, J. O. T. 1988. A microcomputer program for predicting embryonic development in Pacific salmon and steelhead trout. *World Aquacult.* **19**: 80–81.
- Lasker, R., and Theilacker, G. H. 1962. Oxygen consumption and osmoregulation by single Pacific sardine eggs and larvae (*Sardinops caerulea*). *J. Cons. Cons. Int. Explor. Mer.* **27**: 25–33.
- Leatherland, J. F., and Lin, L. 1975. Activity of the pituitary gland in embryo and larval stages of coho salmon, *Oncorhynchus kisutch*. *Can. J. Zool.* **53**: 297–310.
- Li, X., Jensen, E., and Fyhn, H. J. 1989. Effects of salinity on egg swelling in Atlantic salmon (*Salmo salar*). *Aquaculture*, **76**: 317–334.
- Morgan, J. D., and Iwama, G. K. 1991. Effects of salinity on growth, metabolism and ion regulation in juvenile rainbow and steelhead trout (*Oncorhynchus mykiss*) and fall chinook salmon (*Oncorhynchus tshawytscha*). *Can. J. Fish. Aquat. Sci.* **48**: 2083–2094.
- Murray, C. B., and McPhail, J. D. 1988. Effect of incubation temperature on the development of five species of Pacific salmon (*Oncorhynchus*) embryos and alevins. *Can. J. Zool.* **66**: 266–273.
- Peterson, R. H. 1984. Influence of varying pH and some inorganic cations on the perivitelline potential of eggs of Atlantic salmon (*Salmo salar*). *Can. J. Fish. Aquat. Sci.* **41**: 1066–1069.
- Peterson, R. H., and Martin-Robichaud, D. J. 1987. Permeability of the isolated Atlantic salmon (*Salmo salar*) chorion to ions as estimated by diffusion potentials. *Can. J. Fish. Aquat. Sci.* **44**: 1635–1639.
- Potts, W. T. W., and Rudy, P. P. 1969. Water balance in the eggs of the Atlantic salmon, *Salmo salar*. *J. Exp. Biol.* **50**: 223–237.
- Rankin, D. P. 1979. The influence of un-ionized ammonia on the long-term survival of sockeye salmon eggs. *Tech. Rep. Fish. Mar. Serv. Can. No.* 912.
- Rombough, P. J. 1985. Initial egg weight, time to maximum alevin wet weight, and optimal ponding times for chinook salmon (*Oncorhynchus tshawytscha*). *Can. J. Fish. Aquat. Sci.* **42**: 287–291.
- Rombough, P. J. 1986. Mathematical model for predicting the dissolved oxygen requirements of steelhead (*Salmo gairdneri*) embryos and alevins in hatchery incubators. *Aquaculture*, **59**: 119–137.
- Rombough, P. J. 1988a. Aerobic metabolism and dissolved oxygen requirements of embryos and alevins of chinook salmon (*Oncorhynchus tshawytscha*). In *Energy transformations in cells and animals. Proceedings of the Tenth Conference of the European Society for Comparative Physiology and Biochemistry*, Innsbruck, Austria, September 5–9, 1988. Edited by W. Wieser, E. Gnaiger, and L. Poder. University of Innsbruck, Innsbruck, Austria. p. 95.
- Rombough, P. J. 1988b. Growth, aerobic metabolism, and dissolved oxygen requirements of embryos and alevins of steelhead, *Salmo gairdneri*. *Can. J. Zool.* **66**: 651–660.
- Rombough, P. J. 1988c. Respiratory gas exchange, aerobic metabolism, and effects of hypoxia during early life. In *Fish physiology*. Vol. 11A. Edited by W. S. Hoar and D. J. Randall. Academic Press, San Diego. pp. 59–161.
- Rudy, P. P., and Potts, W. T. W. 1969. Sodium balance in the eggs of the Atlantic salmon, *Salmo salar*. *J. Exp. Biol.* **50**: 239–246.
- Saunders, J. W. 1966. Estuarine spawning of Atlantic salmon. *J. Fish. Res. Board Can.* **23**: 1803–1804.
- Shen, A. C. Y., and Leatherland, J. F. 1978. Effect of ambient salinity on ionic and osmotic regulation of eggs, larvae, and alevins of rainbow trout (*Salmo gairdneri*). *Can. J. Zool.* **56**: 571–577.
- Shumway, D. L., Warren, C. E., and Doudoroff, P. 1964. Influence of oxygen concentration and water movement on the growth of steelhead trout and coho salmon embryos. *Trans. Am. Fish. Soc.* **93**: 342–356.
- Silver, S. J., Warren, C. E., and Doudoroff, P. 1963. Dissolved oxygen requirements of developing steelhead trout and chinook salmon embryos at different water velocities. *Trans. Am. Fish. Soc.* **92**: 327–343.
- Steel, R. G. D., and Torrie, J. H. 1980. Principles and procedures of statistics. 2nd ed. McGraw-Hill, New York.
- Veltishcheva, I. F. 1983. Influence of salinity on some species of fish at early stages of development. In *Physiological basis of reproduction of marine and anadromous fish*. Edited by I. A. Shekhanova, L. B. Klyashtorin, and E. A. Kamenskaya. All-Union Research Institute of Marine Fisheries and Oceanography, Moscow. pp. 44–52.
- von Westernhagen, H. 1988. Sublethal effects of pollutants on fish eggs and larvae. In *Fish physiology*. Vol. 11A. Edited by W. S. Hoar and D. J. Randall. Academic Press, San Diego. pp. 253–346.
- Walsh, W. A., and Lund, W. A. 1989. Oxygen consumption by the eggs of the guppy, *Myoxocephalus aeneus*, and the longhorn sculpin, *Myoxocephalus octodecemspinosus*. *Can. J. Zool.* **67**: 1613–1619.
- Walsh, W. A., Swanson, C., Lee, C.-S., Banno, J. E., and Eda, H. 1989. Oxygen consumption by eggs and larvae of striped mullet, *Mugil cephalus*, in relation to development, salinity and temperature. *J. Fish Biol.* **35**: 347–358.
- Weisbart, M. W. 1968. Osmotic and ionic regulation in embryos, alevins, and fry of the five species of Pacific salmon. *Can. J. Zool.* **46**: 385–397.
- Wieser, W., Platzer, U., and Hinterleitner, S. 1985. Anaerobic and aerobic energy production of young rainbow trout (*Salmo gairdneri*) during and after bursts of activity. *J. Comp. Physiol. B*, **155**: 485–492.
- Wieser, W., Forstner, H., Medgyesy, N., and Hinterleitner, S. 1988. To switch or not to switch: partitioning of energy between growth and activity in larval cyprinids. *Funct. Ecol.* **2**: 499–507.
- Wilkinson, L. 1988. SYSTAT: the system for statistics. SYSTAT Inc., Evanston, Ill.